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SEPM - ASSIGNMENT ①

Q.) Explain different software Process Models.

→ software Development life cycle is a simplified representation of a software process. It consists of the following Process models :-

1) * The Waterfall Model -

This is the classic, traditional model where development flows steadily downwards through clearly defined phases. One phase must be completed and 'signed off' before the next begins.

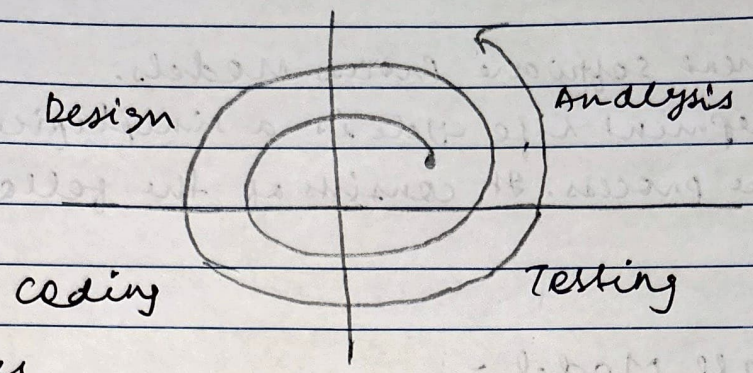
Communication → Planning → Modelling → Construction → Deployment

· Advantages -
simple to understand, strict discipline, clear milestones.

· Disadvantages -
Inflexible to changes; working software is produced late, high risk if requirements are misunderstood early on.

2) * The Spiral Model -

It combines iterative development with systematic controls, moving in 'loops' rather than a straight line. Its primary focus is risk analysis. In every loop you assess potential risks before writing code.



Phases

- Analysis :- Determine objectives, alternatives and constraints.

- Design : Create a blueprint for the software architecture.

- Coding: Develop and code the product or the application.

- Testing: Evaluate the software, deploy it and monitor it for maintainance.

3)* The Incremental Model :-

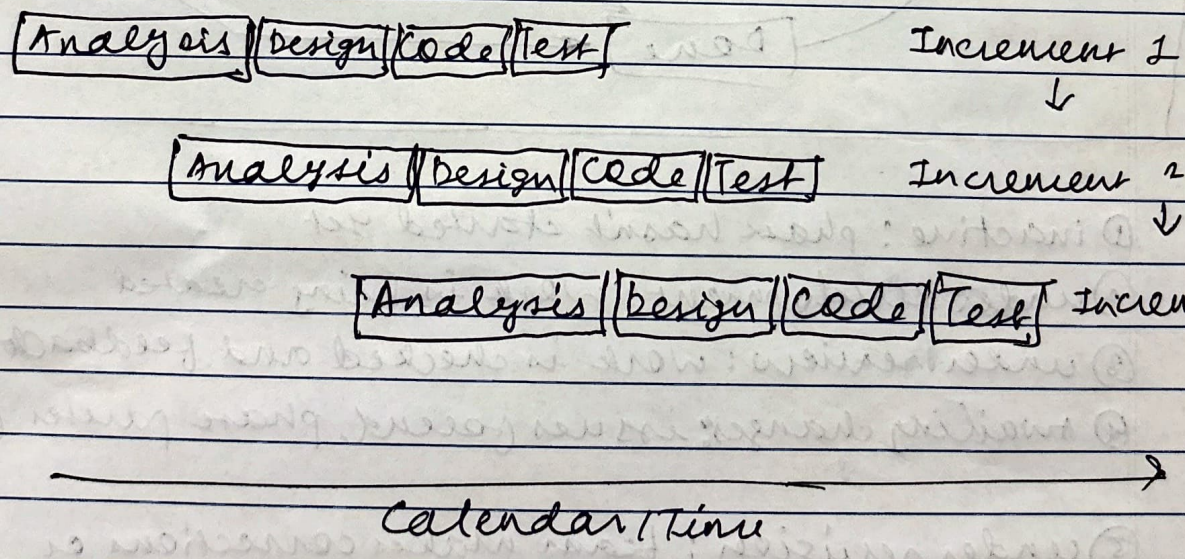
The incremental model is a software process model where the system is developed and delivered in small managable increments. Each increment adds new functionality to the previous release.

Phases in each instrument:
1) Requirement analysis
2) Design
3) Implementation

4) Testing
5) Deployment

Increment 1 → core Features
Increment 2 → Additional features
Increment 3 → Advanced features
⋮
Increment n → Final system

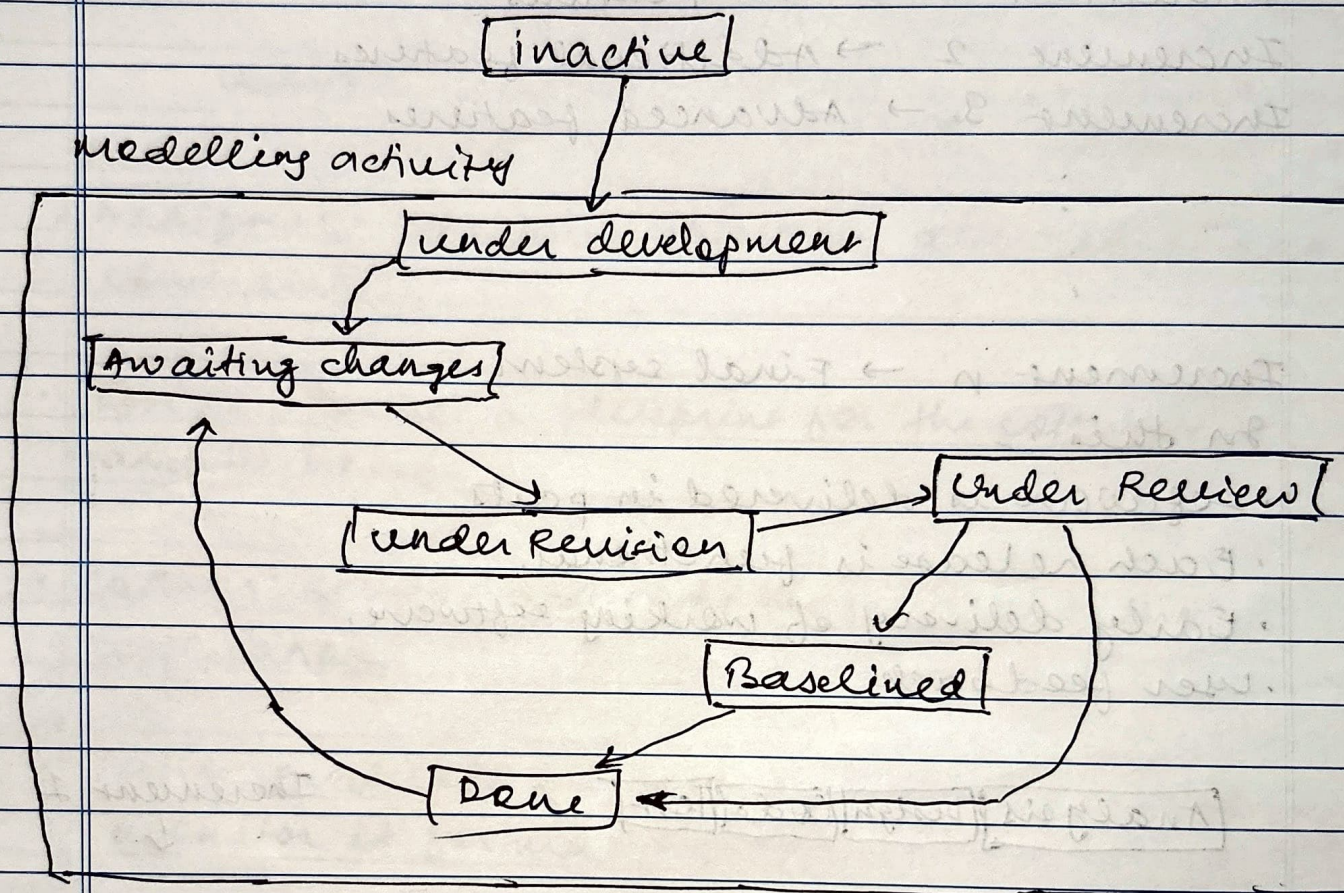
In this :-
- software is delivered in parts.
- Each release is functional.
- Early delivery of working software.
- user feedback



4) * Concurrent Model -

This model allows all SDLC activities to run parallely. Each activity is one of several states. As soon as something changes in one activity

the states of other activities also change. This makes the model flexible and ideal for project where requirements evolve continuously.



① inactive: phase hasn't started yet
② under development: work is being created
③ under review: work is checked and feedback is given
④ Awaiting changes: issues found, phase paused for updates.

⑤ under revision: team makes corrections or modifications

⑥ Baselined: accepted and temporarily frozen.
⑦ done: may reopen if needed.